

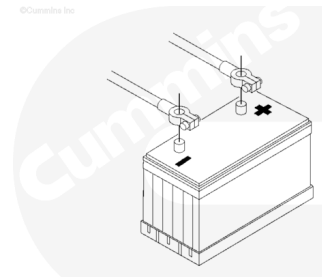
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

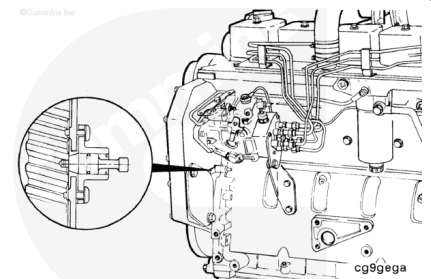
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



Note : The timing pin is used to accurately locate TDC for setting the overhead. The timing pin is typically located below the fuel pump.

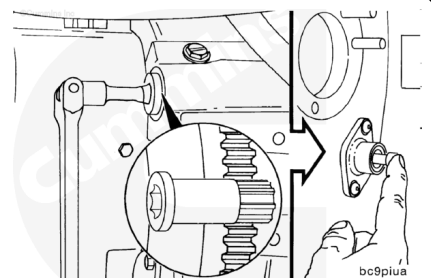
- for front gear train engines, in the front gear housing (shown)
- for rear gear train engines, in the rear gear housing (not shown)



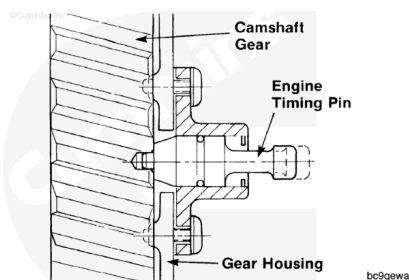
Locate top dead center (TDC) for cylinder number 1 by barring the crankshaft slowly while pressing on the engine timing pin. Barring the engine is recommended from the flywheel on the rear of the engine.

Note : Engine coolant temperature must be less than 60°C [140°F].

Using the barring tool Part Number 3824591, rotate the crankshaft slowly while pressing on the engine timing pin to locate TDC for cylinder number 1.

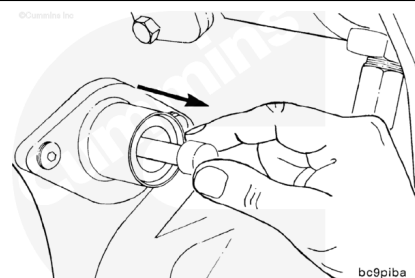


When the timing pin engages in the hole in the camshaft gear, cylinder number 1 is at TDC on the compression stroke.



⚠ CAUTION ⚠

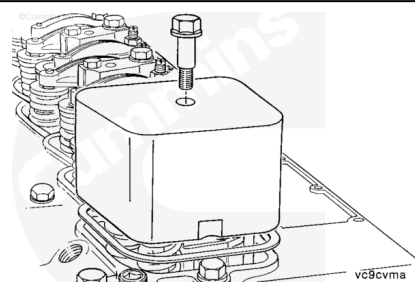
To reduce the possibility of engine or timing pin damage, you must disengage the timing pin after locating top dead center.



Remove

B3.9, B5.9, and B4.5 Engines

Remove the capscrews, capscrew seals, rocker lever covers, and rocker lever cover gaskets.

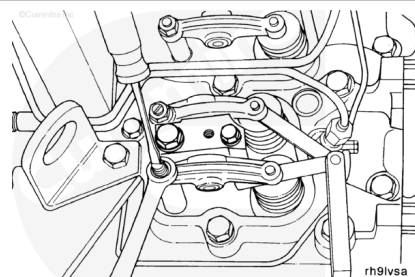


Adjust

B3.9, B5.9, and B4.5 Engines

Note : The clearance is correct when some resistance is “felt” when the feeler gauge is slipped between the valve stem and the rocker lever.

Note : Caution must be used when setting the exhaust valve lash on marine cylinder heads with rotators. The top of the valve stem is slightly recessed below the top of the valve rotator.



Measurements

	mm	in
Intake Clearance:	0.254	0.010

Measurements

	mm	in
Exhaust Clearance:	0.508	0.020

Four-Cylinder Engine Adjustment

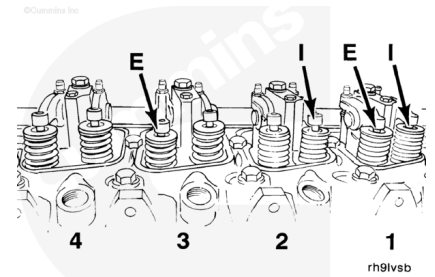
Make sure the engine is at top dead center (TDC) for cylinder number 1.

Set **only** valves indicated by the arrows (E = exhaust, I = intake). Do **not** set valves that are **not** indicated.

Holding the locknut steady with the wrench, adjust the valve clearance with the screwdriver or Allen wrench.

Tighten the locknut and measure the valve lash again.

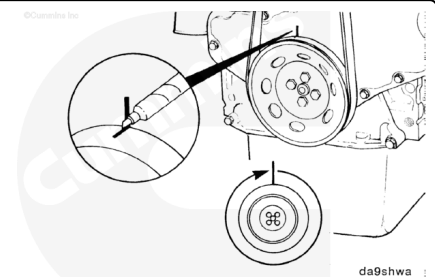
Torque Value: 24 n•m [18 ft-lb]



⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure the timing pin is disengaged.

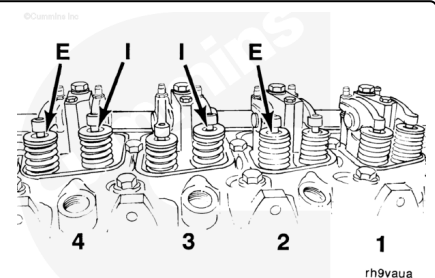
Mark the vibration damper and rotate the crankshaft 360 degrees.



Adjust the valves as indicated in the illustration.

Torque Value: 24 n•m [18 ft-lb]

Set **only** valves indicated by the arrows (E = exhaust, I = intake). Do **not** set valves that are **not** indicated.



Six-Cylinder Engine Valve Adjustment

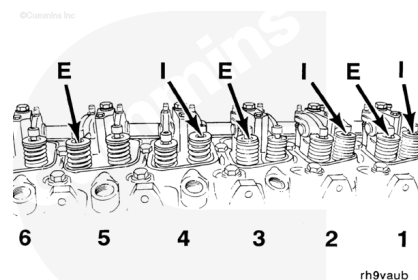
Make sure the engine is at top dead center (TDC) for cylinder number 1.

Set **only** the valves indicated by the arrows in the illustration (E = exhaust, I = intake).

Holding the locknut steady with the wrench, adjust the valve clearance with the screwdriver or Allen wrench.

Tighten the locknut, and measure the valve lash again.

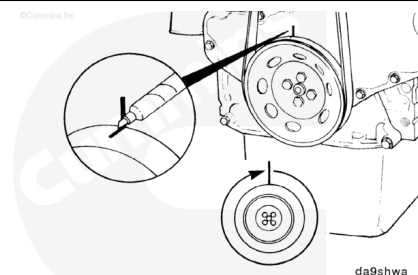
Torque Value: 24 n•m [18 ft-lb]



⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure timing pin is disengaged.

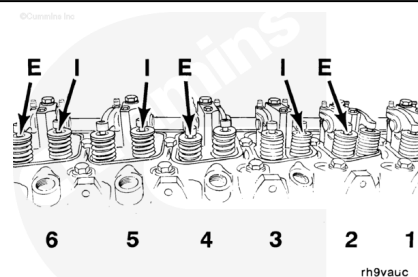
Mark the pulley, and rotate the crankshaft 360 degrees.



Adjust the valves as indicated in the illustration.

Set **only** the valves indicated by the arrows in the illustration (E = exhaust, I = intake). Do **not** set valves that are **not** indicated.

Torque Value: 24 n•m [18 ft-lb]



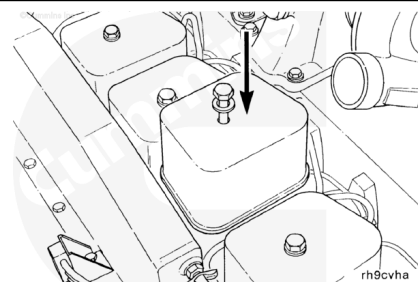
Install

B3.9, B5.9, and B4.5 Engines

Install rocker lever covers with new rocker lever cover gaskets. Next install the capscrews with new capscrew seals.

Tighten the rocker lever cover capscrews.

Torque Value: 24 mm [18 ft-lb]

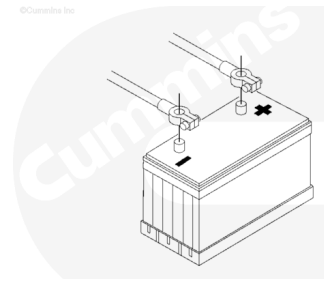


Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries
- Operate the engine and check for leaks.



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